

APM Project Management Qualification

Learner Study Pack



Change Control

Introduction

This section of the Learner Study Pack will cover the Change Control competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of how to manage change requests to alter baseline scope and evaluate them to make a recommendation to approve, reject or defer.

Learning objective and outcomes

Learning objective:

24. Understand change control as the ability to manage variations and change requests in a controlled way.

By the end of this activity, you should be able to:

24a) Understand the purpose and importance of each stage of a typical change control process (such as request, initial evaluation, detailed evaluation, recommendation, update plans and implement)

And

Understand where these stages are different for linear and iterative life cycles.

24b) Know what should be captured and recorded in change requests.

24c) Know ways to assess options related to a proposed change and the high-level impact of the proposed change.

24d) Understand how to justify recommendations about whether to approve, reject or defer changes.

24e) Understand the importance of updating plans and schedules to reflect and communicate changes.

Study time

This section of the Learner Study Pack will take approximately 2 - 3 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Assurance:

- APM Body of Knowledge Seventh edition, 4.3.6 (Change control)
- APM Learning Module, Change control: keeping your projects on track
Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Learning Module, Information sheet: Change control and change request form
Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Resource, What is change control?
Accessed at: <https://www.apm.org.uk/resources/what-is-project-management/what-is-change-control/>
- APM, Planning, Scheduling, Monitoring and Control - The Practical Project Management of Time, Cost and Risk
Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
Accessed at: <https://www.apm.org.uk/book-shop/planning-scheduling-monitoring-and-control-the-practical-project-management-of-time-cost-and-risk/>

Introduction

Change control is a pivotal process that ensures any alterations to project scope, timelines, or resources are managed systematically.

This essential mechanism helps mitigate risks, maintain project stability, and enhance overall project success.

By establishing a structured framework for assessing, approving, and implementing changes, project managers can navigate unforeseen challenges while maintaining control over project objectives.

Effective change control is instrumental in promoting transparency, minimizing disruptions, and fostering adaptability within the dynamic landscape of project development.

Change Control for Linear and Iterative Life Cycles

Change control manages all requests to change the baseline of a project.

These requests can come from different sources for different reasons, ultimately once identified, they need to be evaluated and either approved, rejected or deferred.

Managing change requests (CRs) in a structured way enables the project sponsor and stakeholders to understand the impact on the outcomes of baselined work and influence the decision on how to respond, considering objectives and risk appetite.

The consequence of unmanaged change can be far-reaching within the project environment and to business-as-usual activities.

A retrospective process can be adopted if change is implemented without formal agreement, enabling project documentation and plans to be updated and realistic forecasts made, moving forwards.

In some situations, a change freeze may be needed. This prevents any further change be considered in the project, to protect delivery of the objective.

Projects using an iterative life cycle embed change control as part of the development process, starting each iteration with a planning meeting that clarifies the goals and items to be delivered.

You will still raise CRs, however change is much more normal and expected, therefore it is discussed as part of these planning meetings, rather than needing an exceptional meeting or process to resolve. These may be changes to existing features or the goals and/or items in that iteration, but they are considered alongside the existing requirements.

Change Control Process

The typical stages of a change control process are:

1. Request

CRs can be made by internal or external stakeholders to the project, documenting relevant information on the nature of the change and reasoning for it.

The request should be logged onto the change register, given a unique identifier and assigned an owner to evaluate.

2. Initial evaluation

Considers whether the change should be evaluated in detail or rejected at this stage, looking to determine its necessity, feasibility, potential impact to the project and the rationale for raising.

3. Detailed evaluation

Considers the impact on the project baseline success criteria important to achieving the business case and impacts on work already completed.

Other projects and business as usual should also be considered.

4. Recommendation:

The results of the evaluation are presented to the sponsor and / or wider governance board for a decision, options are:

- Defer – further clarification or information needed
- Reject – cost or risk of the change exceeds the value the change will bring or go beyond agreed thresholds
- Approve – the value being created from the change is worth the cost of implementing

The sponsor is accountable for making sure the decision is communicated.

5. Update plans and implement including communicating the change

The status of the CR is updated in the change log. If the decision is to approve the change, baselined plans and budgets are updated, including the business case if required, and any new risks raised are added to the risk register.

Actions are then implemented in line with the revised plan and monitored through to completion.

Configuration records are updated accordingly to reflect the latest version of project documents incorporating the change and all updated roles and responsibilities, documentation and plans are communicated to stakeholders.

Change Request

To support project evaluations and decisions, the information captured in a change request form typically includes:

- Project overview.
- Description covering the nature of the change.
- Reason for the change.
- Areas impacted.
- Source of change.
- Change request owner.
- Date raised and unique identifier.
- Status of the change request.
- Decision, date made and justification.

Assessing Options and Impact

Options are assessed against the impact they have on achieving the success criteria set out in the business case, and whether the skills and resources needed to deliver the change are available.

This essentially comes back to feasibility and practicality.

A change may impact any of the following:

- Benefits
- Scope
- Schedules
- Costs
- Quality
- Resources
- Risks
- Stakeholders

Understanding why the change has been raised is also important when assessing options – is it mandatory, a change in regulation, risk profile, customer driven or financial constraints driving the request?

Another consideration is the degree of resources that will be consumed to undertake the assessment. Before evaluating, the project should consider whether they should be assessing such a change at all.

Recommendations

A recommendation is driven by looking at whether the proposed change could jeopardise the achievement of the project objectives.

When justifying the recommendation, the following should be considered:

- Which areas of the project will be affected, how will they be affected, and does the project have the skills and resources needed to handle it?
- Does the project have sufficient contingency budget to drawdown, or would further funding need to be requested?
- If additional resources are needed, does the business have the capacity and the capability to source them, or would external (to the project) help be needed?
- Does it affect the end date, or can the project maintain delivery on time? If they can't, will the project outcomes still be of use to the business?
- Are the benefits listed in the business case going to be materially impacted in any way, if so how and what does that mean both for the project and the business?

Updating Plans and Schedules to Reflect Changes

One of the most common reasons why projects fail is poorly managed change.

Managing requests for change effectively is a proven success factor for project delivery.

The project management plan sets out what is expected from the project team. If this is not updated and communicated when changes are made, the team will be working off outdated information, costing the project time and money.

Causing unnecessary conflict amongst the team on what needs to be delivered, impacting team performance and satisfaction levels.

The solution could end up redundant if what is delivered is not fit for purpose or not delivered in time.

Stakeholders could disengage through the loss in confidence of the project team's ability to deliver the project's outcomes to the agreed success criteria.

Could have a knock-on impact on other change activity happening in the business or to business-as-usual activity.

Ultimately, the benefits in the business case may not be realised, impacting the return-on-investment.

Scenario One:

Ayo, the Founder/Director, is an incredibly creative and energetic person. They feel that changes can be made at any time to anything!

Understandably this has made you rather nervous when it comes to the nature of a complex renovation like the one planned at 'Snappy Appy'.

Knowing that Ayo could initiate changes, you must ensure that a robust change control process is put into place.

How would you approach the different stages in the change control process given the linear nature of the project?

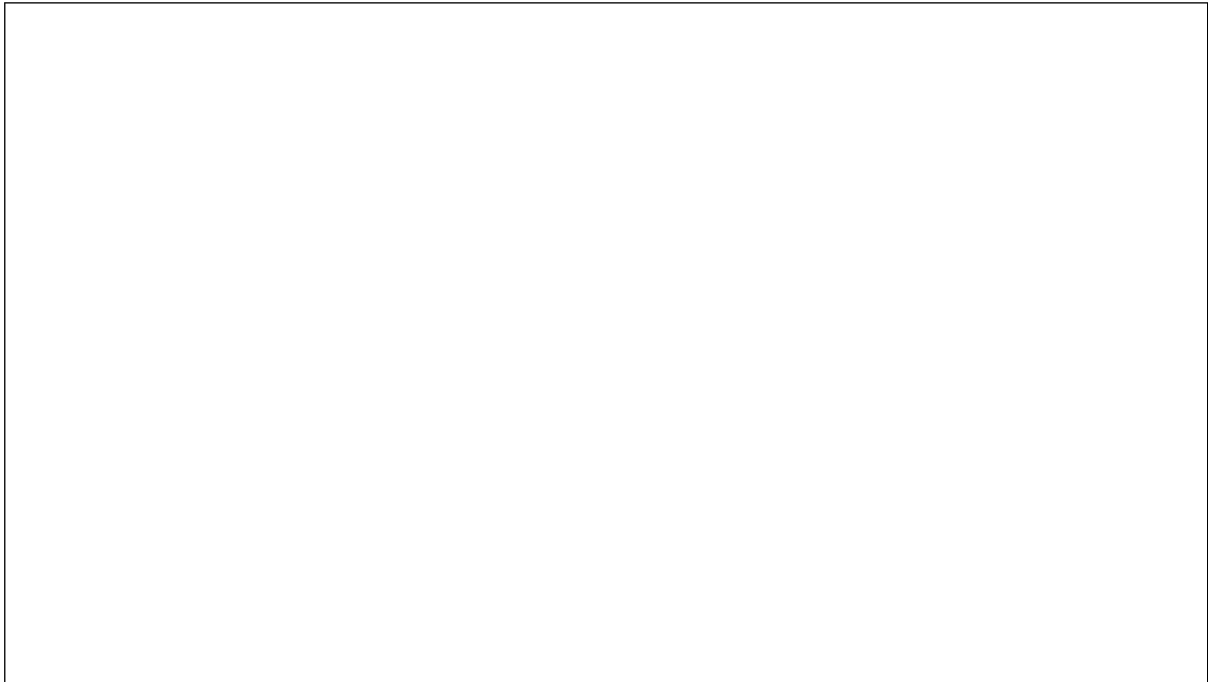
How would this be different when dealing with an iterative project life cycle?

Scenario Two:

As part of your change control process, the collection of accurate, practical and useful information is paramount.

With the potential for multiple stakeholders to submit change requests throughout the project, you must ensure that the change request is clear and concise.

Draft a change request form for the 'Snappy Appy' renovation project that includes all the required information to support project evaluations and decision making.

A large, empty rectangular box with a thin black border, intended for the student to draft a change request form for the 'Snappy Appy' renovation project.

Scenario Three:

Ayo, the Founder, and Lee, the HR Manager, have been discussing the functionality of some of the spaces in the original renovation plan and would like to make some changes.

The original plan includes a brand-new conference room designed to allow up to 18 people to meet face to face as a team.

In addition, three video call pods will be installed to allow one to one video conferencing calls. These pods will be accessible to all members of staff and will be reserved using an online booking system.

However, the Customer Services Team works on a hybrid basis with at least 2 members working solely from home. This means that they will be unable to attend team meetings unless video conferencing functionality is installed in the conference room.

After Lee explained this to Ayo, they agreed that the changes need to be made to the conference room to allow it to be used by all teams.

What impact could the changes have on the delivery of the overall project, e.g. cost, timeframes etc?

Which plans or schedules may have to be updated as a result of the proposed change?

If these plans or schedules are not updated, what could be the potential impact on the project?

Scenario Four:

Robin, the Finance Manager, has heard about the proposed changes to the conference room and is concerned about the impact that these changes will have on the renovation project.

They believe that the impact of one team not being able to meet together doesn't sound like an issue that should involve changing the design of such a significant part of the build.

What additional information or questions could you use to explore whether the recommendation should be approved or not?

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Change Control section:

- How could you communicate revised project plans to ensure that changes are understood?
- What are the potential reasons why a change could be deferred?
- What are examples of reasons why stakeholders might oppose a change control process?
- When might it be appropriate to instil a change freeze in a project?

Exam practice questions

24.1 In a change control process, what is considered in the 'Detailed Evaluation' stage?
(1 mark)

- a. The decision to approve, defer or reject the change.
- b. The updates required to the change log.
- c. The communication of the change to stakeholders.
- d. The impact on the project's business case.

24.2 Which piece of information is not required during the initial 'Raise Change Request' stage of the Change Control process?
(1 mark)

- a. A high-level summary of the change.
- b. Who has requested the change.
- c. A justification for the change being requested.
- d. A detailed evaluation of the change's impact on the project.

24.3 You are managing a construction project that involves building a residential housing estate. During the project implementation phase, a key stakeholder suggests adding a rooftop garden to enhance the overall appeal of the complex.

The stakeholder believes it would be a valuable addition and will increase the property's market value. The project is currently on schedule and within budget.

As the project manager, what is the most appropriate action to take?
(1 mark)

- a. Log the request for the rooftop garden, accept the suggestion as it will deliver additional benefits and inform the stakeholder.
- b. Reject the rooftop garden suggestion to avoid any scope changes, and inform the stakeholder.
- c. Log the request for the rooftop garden, assess the impact of the proposed change and discuss with the stakeholder.
- d. Log the request for the rooftop garden and inform the stakeholder of the deferral to avoid disruptions.

24.4 A change is requested on your project. You have to either approve, reject or defer this request.

Which of the below would be the main reason to defer?
(1 mark)

- a. The cost impact would be too high.
- b. The time impact would be too long.
- c. There is insufficient information.
- d. The cost has no impact.

24.5 A change request has been approved that impacts the project's business case. What are the Project Manager's next steps?
(1 mark)

1. Update baselines and budgets including the project benefits.
2. Update the Sponsor of the decision and request they raise any new risks because of the change.
3. Communicate to stakeholders and update roles and responsibilities to implement the change.
4. Update configuration records and latest documentation to reflect the change.
5. Communicate to the project team so they can decide what needs to be updated.
6. Communicate to the sponsor and defer the deployment to a later stage.

Select the correct combination of statements:

- a) 1, 3, 4
- b) 1, 4, 5
- c) 2, 3, 4
- d) 4, 5, 6

24.6 You are managing a project in a linear lifecycle. A change request has been raised.

- (a) Explain the importance of change control in a linear lifecycle. (4 marks)
- (b) Following a detailed evaluation, state the 3 possible recommendations arising from a change request. (1 mark)

24.7 Within a Change Control Process:

- State two impacts that a change request can have on a project and why it is important to evaluate those impacts. (4 marks)
- State two recommendations a project team can provide for a change request (1 mark).

24.8 What are two pieces of information that should be captured and recorded in a change request?
(2 marks)

24.9 You are the Project Manager overseeing the development of a new software system to manage the company's payroll. The project is part of a larger programme, and as such, it shares a common budget, resources, and timeline with other projects.

As part of your project's change control process, a change has been approved that fundamentally alters the scope.

What are two potential impacts on the programme if you do not adequately communicate this approved change?
(2 marks)

24.10 You are managing a project to implement a new software system for your organisation. You are 18 weeks into the implementation phase, and a change request has been submitted that proposes adding additional functionality to the solution. You progress the change request through the project's change control process.

Alongside the team, you complete the detailed evaluation of the change request, confirming that the ____ (a) ____.

Next, you take the change to your Project Sponsor for approval. During this approval meeting they will either accept, ____ (b) ____ the change.

(2 marks)

(a)

1. Change's value is worth the cost
2. Communication strategy is clear
3. Project's plan has been updated
4. Configuration record has changed

(b)

1. Escalate or reject
2. Investigate or reject
3. Escalate or defer
4. Reject or defer

Exam practice answers

Question	Answer / Mark scheme						
24.1	D						
24.2	D						
24.3	A						
24.4	C						
24.5	A						
24.6	(a) 4 marks. 1 mark, up to a max of 4, for any of the following: 1. Identifying linear projects are scope and quality fixed (rather than iterative which tend to be time and cost fixed). 2. Ensures changes to scope/requirements/deliverables are made in a controlled way with appropriate assessment 3. Ensures alignment of agreed changes with project objectives/business case 4. Controls scope creep 5. Stating the project could go over budget and/or over schedule due to uncontrolled change. 6. Supports management of risks and impacts on dependencies elsewhere in the project by having a controlled process 7. Ensures effective communication with stakeholders to ensure transparency, accountability and robust decision making 8. Provides an audit trail of decisions on proposed changes throughout the course of the project (b) 1 mark total for Approve (Accept), reject, defer.						
24.7	1 mark for each correct impact stated, 2 marks available in total. Any of the below or other correct answer: • Impact on project benefits • Impact on project risk • Impact on project budget • Impact on project resources • Impact on project timeline / schedule • Impact on the business case 1 mark for each correct explanation of why it is important to evaluate those impacts, 2 marks available in total. Any of the below or other correct explanation: <table border="1"> <tr> <td>Impact on project benefits</td><td> • The change may deliver additional benefits. • The change may reduce or change existing benefits. </td></tr> <tr> <td>Impact on project risk</td><td> • The change could impact the project risks. • The change could go over the project risk tolerance. • The change could create new project risks. </td></tr> <tr> <td>Impact on project budget</td><td> • The change could require additional budget. • The change could go over the project contingency funds. </td></tr> </table>	Impact on project benefits	• The change may deliver additional benefits. • The change may reduce or change existing benefits.	Impact on project risk	• The change could impact the project risks. • The change could go over the project risk tolerance. • The change could create new project risks.	Impact on project budget	• The change could require additional budget. • The change could go over the project contingency funds.
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	<table border="1"> <tr> <td data-bbox="352 203 683 360">Impact on project resources</td><td data-bbox="683 203 1380 360"> - The change could require additional team members / external contractors / equipment / supplies. - The change could impact the capacity of the project team – either increasing or decreasing. </td></tr> <tr> <td data-bbox="352 360 683 427">Impact on project timeline / schedule</td><td data-bbox="683 360 1380 427"> - If the change involves more work, the timeline / schedule may need to be re-planned. </td></tr> <tr> <td data-bbox="352 427 683 495">Impact on business case</td><td data-bbox="683 427 1380 495"> - If the change alters the business case, it could mean the project is no longer viable. </td></tr> </table> 1 mark for any of the below, 2 marks available in total. Only 2 recommendations need to be stated to gain the marks: - Approve / Accept - Reject - Defer	Impact on project resources	- The change could require additional team members / external contractors / equipment / supplies. - The change could impact the capacity of the project team – either increasing or decreasing.	Impact on project timeline / schedule	If the change involves more work, the timeline / schedule may need to be re-planned.	Impact on business case	If the change alters the business case, it could mean the project is no longer viable.
Impact on project resources	- The change could require additional team members / external contractors / equipment / supplies. - The change could impact the capacity of the project team – either increasing or decreasing.						
Impact on project timeline / schedule	If the change involves more work, the timeline / schedule may need to be re-planned.						
Impact on business case	If the change alters the business case, it could mean the project is no longer viable.						
24.8	1 mark for each correct piece of information, 2 marks available in total – any of the below: - Project overview - Description of the change - Rationale / reason for the change - Source of change / who has raised the change / Change request owner - Date raised - Unique identifier - Status - Initial evaluation - Detailed evaluation / Impact analysis - Decision / Recommendation - Date of decision / recommendation - Priority - Approval level - Proposed solution - Follow-up actions						
24.9	1 mark per impact, 2 marks available in total – any of the below: - The programme may go over budget if the change to the project requires additional budget. - The resource of the programme might be impacted, if the change to the project needs more resource. - The timelines of the programme may need to be re-planned / may be changed / may be delayed. - The business case for the programme may no longer be valid / need to be changed. - There may be impacts in other areas of the programme as a result of this change						
24.10	(a) 1 (b) 4						